

FLAGSHIP SOLUTION SCALING TRACK PROFILE

Commercialization of Climate-Resilient, Nutrient-Dense Beans in Kenya & Malawi

Scaling for Impact Science Program | Area of Work 2

Grand Challenge 2 · Deliver Inputs and Services at Scale through Inclusive Market Systems

Flagship 2.1 · Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

Version 1.0 · June 2026



Foreword

CGIAR SCALING FOR IMPACT · AREA OF WORK 2 — PATHWAYS TO SCALE IN AGRIFOOD SYSTEMS

CGIAR produces far more scale-ready innovation than currently reaches farmers at scale. Closing that gap is the purpose of the CGIAR Scaling for Impact (S4I) Science Program. Within S4I, Area of Work 2 — Pathways to Scale in Agrifood Systems — is organized around three Grand Challenges that mark where scaling can unlock the largest development returns across food, land, and water systems in a climate crisis. These Grand Challenges were defined through an evidence-led process — drawing on the CGIAR Impact Report 2022–2024, Initiative reporting, and an analysis of eleven major donor strategies — so that each rests on demonstrated scaling potential and connects directly to the priorities of CGIAR's funders.

Under each Grand Challenge sits a focused set of **Scaling Flagships** that "dock with" and support CGIAR's Science Programs — taking mature innovation bundles and accelerating their delivery and scaling systems:

- **Grand Challenge 1 — Close the Climate Adaptation Gap** through scalable climate services and smart water management

Flagship 1.1 — Digital Climate Services for Resilient Farm & Enterprise Management

Flagship 1.2 — Scaling Smart Irrigation & Climate-Resilient Water Systems

- **Grand Challenge 2 — Deliver Inputs and Services at Scale** through inclusive market systems

Flagship 2.1 — Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

Flagship 2.2 — Entrepreneurship through Mechanization & Postharvest Services

Flagship 2.3 — Genetics, Inputs & Services for Animals & Aquaculture

- **Grand Challenge 3 — Drive Demand, Shape Behavior & Activate Policy** for nutrition

Flagship 3.1 — Catalyzing Demand for Nutritious Diets through Markets, Institutions & Policy

Each Flagship is delivered through **Scaling Solution Tracks** — the profile that follows is one of them. Tracks are run deliberately as large, time-bound scaling campaigns, not as research projects or pilots. Experience across the portfolio shows that the binding constraint to scaling is rarely the innovation itself; it is *system integration* — the delivery and business models, finance and risk-sharing, institutional and policy alignment, and coordination among actors required to move a proven solution into widespread use. Each track therefore bundles a validated innovation with a clear delivery pathway, committed partners, and a defined geography, and is driven through and with scaling partners rather than by CGIAR alone — leveraging complementary investments to amplify reach.

This campaign approach lets S4I concentrate resources, operate in performance-based mode against measurable KPIs — farmers adopting innovations, jobs created, consumers gaining access to healthier diets, and scaling finance and supportive policy mobilized — and hold itself accountable to the S4I 2030 outcome targets and CGIAR's Impact Area goals. Across every track, scaling is pursued responsibly and inclusively, with women, youth, and marginalized groups systematically included by design.

1. Overview

In Kenya and Malawi, common bean is a critical staple and source of income for millions of smallholder farmers. Yet decades of breeding for climate resilience and enhanced nutrition remain stifled by chronically underdeveloped seed systems. Despite the release of over 30 improved varieties, adoption languishes below 10%, with formal seed channels accounting for only 5–10% of seed sourced. The core problem is a systemic failure to connect seed production with market demand — fragmented coordination, poor demand forecasting, and weak last-mile delivery.

This scaling track transforms the bean seed value chain by commercializing a bundle of climate-resilient, nutrient-dense varieties — including **Nyota, Faida, and Angaza** in Kenya, and **NUA45 and SUG131** in Malawi. The goal is a demand-led, market-anchored seed system reaching **100,000 farmers (60% women) on 50,000 hectares by 2028**. The solution is not merely the seed, but an innovation bundle integrating (i) improved genetics and agronomic practices, (ii) inclusive multi-stakeholder platforms (MSPs) for coordination, (iii) digital tools for traceability and demand forecasting, and (iv) innovative financial services to unlock credit for farmers and SMEs.

The initiative builds on the robust foundation of the **Pan-Africa Bean Research Alliance (PABRA)**, whose demand-led model has demonstrated scalable impact across Africa over two decades. With confirmed commitments from AGRA, national research systems (KALRO, DARS), private-sector actors (Spice World Ltd, Equity Bank), and digital innovators (Pesira), this track is poised to move from a well-functioning but incremental model to a systemic, accelerated scaling pathway — creating a replicable blueprint for public–private partnership in the CGIAR.

2. Innovation Bundle & Scaling Ambition

2.1 What is being scaled

Technologies: Certified seeds of climate-resilient, high-iron, high-yielding bean varieties (e.g., Nyota, Faida, Waithera, KK-Rosecco 33, Angaza in Kenya; NUA45, SUG131 in Malawi).

Practices: Good agronomic practices, post-harvest handling, and financial-literacy training bundled with seed access.

Systemic approaches:

- **Multi-stakeholder Platforms (MSPs):** Formalized platforms coordinating seed producers, grain off-takers, financiers, and extension services to align supply with demand.
- **Digital tools:** Farmer profiling and seed traceability systems (via Pesira) for real-time demand forecasting and market transparency.
- **Innovative finance:** Bundled financial services and credit access (via Equity Bank) for farmers and last-mile aggregators.

2.2 Evidence of impact

- **Established model:** PABRA's demand-led seed systems reached over 2.7 million Africans through school-meal programs alone in 2024.
- **Market uptake:** In Malawi, the MSP model increased certified-seed purchases and boosted farmer willingness to pay by 25%.
- **Transferability:** Successfully adapted to other crops (cowpea in Nigeria, groundnuts in Tanzania) and regions (Zambia).
- **Policy influence:** PABRA engagement contributed to seed-policy reforms in Kenya, including the National Seed Policy.

2.3 Scaling targets

Table 1. Scaling targets — 2026 S4I investment vs. long-term goal

Outcome metric	2026 target (S4I investment)	Long-term goal (3–5 years)
Farmers reached	10,000 farmers (60% women, 20% youth)	100,000 farmers (60% women, 20% youth)
Area under improved practices	3,500 hectares	50,000 hectares
Certified seed distributed	600 tons	3,000 tons
Jobs supported	200 youth employment opportunities	2,000 direct/indirect jobs in seed & grain value chains
Investment leveraged	USD 480,000 (AGRA) + private-sector in-kind	70% cost recovery for seed enterprises by 2028

3. Scaling Pathway & Theory of Change

3.1 Pathway strategy

The scaling pathway is a structured public–private partnership that commercializes the bean value chain by anchoring seed production in verified market demand. It addresses the critical bottleneck of weak coordination between seed supply and grain markets, which has historically led to mismatches and disincentivized private investment. The strategy follows a phased approach:

- **Orchestration:** S4I and AGRA funding catalyze digitalized MSPs in key bean corridors, bringing together seed companies, aggregators (e.g., Spice World Ltd), financial institutions (Equity Bank), and farmer groups to co-develop production and procurement plans based on forecasted demand.
- **De-risking & financing:** Linking farmers directly to institutional buyers (school-feeding programs, food processors) and providing credit through Equity Bank de-risks both production and market access. Digital profiling via Pesira supplies the data for accurate forecasting and lending decisions.
- **System integration:** Planning and coordination functions transition to national institutions (KALRO, DARS) and private-sector actors, embedding demand-led coordination into national systems for sustainability beyond the project period.

3.2 Theory of Change

If smallholder farmers — particularly women and youth — gain access to a bundled innovation package (improved seed, finance, digital tools, agronomic training) through inclusive MSPs, and these platforms align seed production with the predictable demand of institutional off-takers and grain processors, **then** farmers adopt improved varieties at scale, raising productivity and incomes — which in turn incentivizes seed companies to invest in multiplication and distribution, **ultimately** building a vibrant, commercially sustainable bean value chain that enhances food security and resilience in Kenya and Malawi.

3.3 Key activities funded by S4I

Mobilization & diagnostics

- Conduct a readiness assessment to map institutional arrangements and finalize entry points.
- Establish and strengthen digitalized MSPs in Kenya and Malawi.

Market & financial linkages

- Initiate joint seed-production and grain-procurement planning with seed companies, off-takers, and farmer groups.
- Deploy digital profiling tools (via Pesira) for real-time demand forecasting and farmer registration.
- Facilitate financial-literacy training and bundled credit services with Equity Bank.

Partnership & capacity building

- Formalize partnerships with six new private-sector and development actors.
- Train 500 service providers (extension agents, agro-dealers) to support last-mile delivery.

4. Partners & Roles

Table 2. Partner types, actors, and roles in the scaling pathway

Partner type	Specific partners & quantities	Role in scaling pathway
Demand partners	Private sector: Spice World Ltd, Delish & Nutri, Chrubet (KE); TAF Foods (MW). Financial: Equity Bank. Farmer orgs: cooperatives and youth/women-led producer groups.	Aggregate demand, provide predictable off-take contracts, deliver credit and financial services to farmers, and serve as last-mile distributors.
Scaling partners	CGIAR: ABC-PABRA, IITA, CIMMYT. Public sector: KALRO (KE), DARS (MW), County Govts (KE), DAES (MW). Development: AGRA (CESSA), BRAINS Project, TAAT Bean Compact. Digital: Pesira.	Provide technical backstopping, breeding expertise, policy alignment, and catalytic co-funding; lead EGS production, certification, and system digitization.

4.4 Funding model: S4I provides catalytic funding for coordination, partnership development, and technical assistance, while AGRA and bilateral projects (e.g., BRAINS) co-finance seed-hub development and market activation. Private-sector partners invest in multiplication and distribution, leveraging licensing revenues and market growth.

5. Enabling Environment & Risks

5.1 Opportunities

- **Policy alignment:** Aligns with the COMESA seed-harmonization policy, Kenya's National Seed Policy (2010), and Malawi's Agriculture Commercialization policy — a favorable regulatory environment.
- **Partner commitment:** AGRA has committed USD 480,000 to co-finance seed-hub development and policy alignment.
- **Private-sector appetite:** Signed MOUs with Spice World Ltd and Equity Bank signal strong interest in traceability, contract farming, and bundled financial services.
- **Proven platform:** The TAAT High-Iron Bean Compact provides a ready-made scaling platform already reaching over 898,000 farmers across Africa.

5.2 Challenges & mitigation

- **Fragmented coordination.** Risk of mismatches between seed supply and market demand. *Mitigation:* joint seed-planning platforms (digitalized MSPs) where all actors co-develop seasonal plans, building on Malawi and Zambia models.
- **Weak demand forecasting.** Inaccurate projections undermine private-sector confidence. *Mitigation:* deploy Pesira to capture real-time demand and anchor planning with off-take agreements from institutional buyers.
- **Partner dependency.** Ambition is "subject to AGRA commitments." *Mitigation:* a focused 2026 workplan delivering foundational outcomes even if AGRA funds are delayed, plus fallback to other bilateral projects (BRAINS, TAAT).
- **Inclusion risks.** Women and youth may be excluded from new market structures. *Mitigation:* GenderUp tool for inclusive design; disaggregated KPIs; targeted training; gender-responsive packaging (small seed packs); representation in all planning forums.

6. Learning & Adaptation

6.1 Monitoring focus

- **Process outcomes:** functionality of digital MSPs, number of joint plans developed, frequency of partner engagement.
- **Behavior change:** adoption of digital tools by farmers and SMEs, shifts in Equity Bank lending, use of demand forecasts by seed companies.
- **System outcomes:** seed-distribution efficiency, reduction in demand–supply mismatches, cost-recovery rates of seed enterprises.
- **Inclusion:** disaggregated participation rates, income changes for women and youth, access to credit.

6.2 Adaptive management

Learning is continuous and drives real-time adaptation through a results-based MELIA framework anchored by:

- **Digital dashboards:** real-time data from Pesira and partner reports enable quarterly progress reviews against targets.
- **Quarterly review meetings:** PABRA and S4I management assess performance, address bottlenecks, and refine strategies based on evidence.
- **Participatory feedback loops:** farmer surveys and aggregator reports ground-truth the model and keep it responsive to user needs.

7. Next Steps & Timeline (2026)

Table 3. 2026 quarterly implementation roadmap

Quarter	Key activities
Q1	Finalize scaling-track profile and partnership MOUs. Conduct readiness assessment and stakeholder mapping. Initiate EGS production planning with KALRO and DARS.
Q2	Formalize six new partnerships. Launch digital MSPs in Kenya and Malawi. Identify seed-hub sites and youth groups. Customize and deploy financial-literacy tools with Equity Bank.
Q3	Train first cohort of 500 service providers. Begin farmer profiling and seed-demand data collection via Pesira. Facilitate first joint planning meetings between seed companies and off-takers.
Q4	Consolidate year-one lessons. Report on 2026 targets (10,000 farmers reached). Refine 2027 workplan based on data. Document the "digital MSP" model for communication.

Communications & knowledge: a 10-minute video profiling the digital-MSP model, quarterly blog posts highlighting partner successes (e.g., Spice World's contract farming), and a policy brief on the role of digital tools in seed-system coordination.

8. Contact

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